Information Collection Strategies Experiment

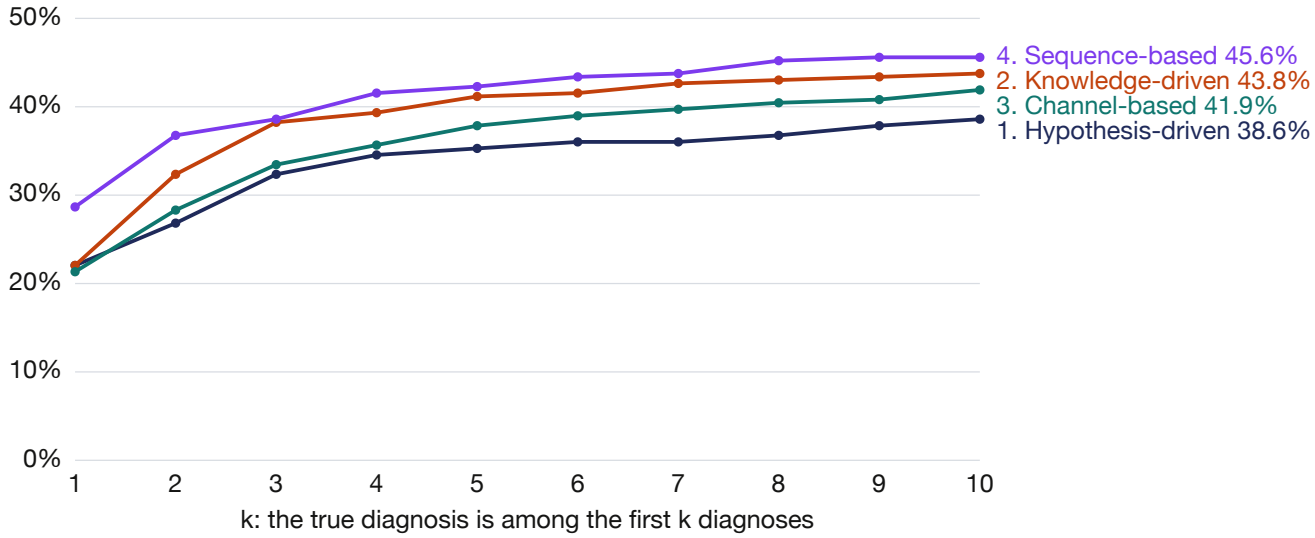
Diagnostic accuracy of four information collection strategies on 272 simulated cases

Run 2026-10-19T17:59:10+00:00 to 2026-10-19T19:40:02+00:00 · record `strategies_272_8it.json` · status completed · commit 06670ca

Result

Strategy	Hit rate@1	Hit rate@3	Hit rate@5	Hit rate@10	nDCG@10	MRR	Cases
1. Hypothesis-driven	22.1%	32.4%	35.3%	38.6%	0.301	0.274	272
2. Knowledge-driven	22.1%	38.2%	41.2%	43.8%	0.335	0.302	272
3. Channel-based	21.3%	33.5%	37.9%	41.9%	0.314	0.281	272
4. Sequence-based	**28.7%**	**38.6%**	**42.3%**	**45.6%**	**0.373**	**0.347**	272

Bold marks the best value in a column. Hit rate@k: share of cases whose true diagnosis code is among the first k codes of the strategy's differential. nDCG@10 and MRR reward a true diagnosis placed higher. Every case counts in the denominator.



How sure the differences are

Strategy	Hit rate@1 (95% interval)	Hit rate@10 (95% interval)
1. Hypothesis-driven	22.1% (17.5% to 27.4%)	38.6% (33.0% to 44.5%)
2. Knowledge-driven	22.1% (17.5% to 27.4%)	43.8% (38.0% to 49.7%)
3. Channel-based	21.3% (16.9% to 26.6%)	41.9% (36.2% to 47.8%)
4. Sequence-based	28.7% (23.6% to 34.3%)	45.6% (39.8% to 51.5%)

Wilson intervals over the cases of this single run. The strategies ran on the same cases, so the paired comparison below is the sharper test.

Pair (first vs second)	@1: cases only the first / only the second hit	p	@10: only the first / only the second	p
1. Hypothesis-driven vs 2. Knowledge-driven	24 / 24	1.000	24 / 38	0.098
1. Hypothesis-driven vs 3. Channel-based	28 / 26	0.892	32 / 41	0.349
1. Hypothesis-driven vs 4. Sequence-based	22 / 40	0.030	23 / 42	0.025
2. Knowledge-driven vs 3. Channel-based	27 / 25	0.890	35 / 30	0.620
2. Knowledge-driven vs 4. Sequence-based	22 / 40	0.030	33 / 38	0.635
3. Channel-based vs 4. Sequence-based	16 / 36	**0.008**	30 / 40	0.282

Exact McNemar test on the same 272 cases. With 6 comparisons, a p-value below 0.008 (bold) holds after a Bonferroni correction; values between that and 0.05 are suggestive only.

What was run

- 272 cases from `cases/combined_272_whole_chunking.json`, each run with every strategy: 1088 case runs.
- Each strategy could ask at most 8 questions and could stop earlier by its own rule. When it stopped, it gave a differential of up to 10 diagnoses with probabilities.
- Strategy model: anthropic/claude-haiku-5.5. Simulated environment model: openai/gpt-6-luna; literature sources europe_pmc, litsense.
- Coding: ICD-11 MMS 2024-01, first result of the flexible search for the diagnosis name. A prediction counts as a hit only when its code equals the true diagnosis code exactly.
- Total time 1.7 h; total cost $12.47 (strategies $5.74, environment $6.72).

How the strategies behaved

Strategy	Questions per case	Why runs stopped	Strategy LLM calls per case	Seconds per case	Strategy cost	Environment cost	Failed runs	No differential
1. Hypothesis-driven	7.62	max iterations 233, strategy done 39	15.4	353	$2.25	$1.66	0	0
2. Knowledge-driven	8.00	max iterations 272	16.0	335	$1.71	$1.67	0	0
3. Channel-based	7.66	max iterations 249, strategy done 23	17.5	351	$0.67	$1.72	0	0
4. Sequence-based	7.53	strategy done 117, max iterations 155	14.5	346	$1.11	$1.68	0	0

Where the answers came from (questions)	Case study	Literature	Unanswerable	No answer (error)
1. Hypothesis-driven	515	1463	94	0
2. Knowledge-driven	632	1459	85	0
3. Channel-based	490	1473	121	0
4. Sequence-based	477	1452	118	0

Coding

Strategy	Predicted diagnoses	Without an ICD code	Cases with no coded prediction	Cases whose true diagnosis has no code
1. Hypothesis-driven	2720	2	0	2
2. Knowledge-driven	2720	1	0	2
3. Channel-based	2720	1	0	2
4. Sequence-based	2720	9	0	2

4362 distinct diagnosis names were coded; 11 have no ICD-11 match. A case whose true diagnosis has no code cannot be hit by any strategy.

Reading these numbers

- Exact code match is strict: a correct diagnosis worded so that the search returns a neighbouring code (a parent, a sibling, or a code with a different extension) counts as a miss.
- One run per case and strategy. LLM sampling differs between runs, so small differences between strategies may not repeat.
- The stopping thresholds are starting values and were not tuned.
- Answers marked "literature" are synthesized by the simulated environment when the case text is silent; they are plausible, not facts about the patient.